

Minseong Kweon

Website: mnseong.github.io

Email: kweon021@umn.edu

Phone: +1 (763) 458-5075

RESEARCH INTERESTS

Neural Rendering, 3D Reconstruction, Multi-View Stereo, Depth Estimation

EDUCATION

University of Minnesota, Twin Cities, United States

Sep. 2025 — present

- M.S. in Robotics

Pusan National University, Busan, Republic of Korea

Mar. 2019 — Feb. 2025

- B.S. in Mechanical Engineering

- Served mandatory military service

Aug. 2020 — May. 2022

PUBLICATIONS

2. NoiseGS: Boosting 3D Gaussian Splatting with Positional Noise for Large-Scale Scene Reconstruction

Minseong Kweon, Kai Cheng, Xuejin Chen, Jinsun Park

- *European Association for Computer Graphics (Eurographics)* short paper, 2025. [\[Paper\]](#) [\[Project Page\]](#)

1. ULTRON: Unifying Local Transformer and Convolution for Large-Scale Image Retrieval

Minseong Kweon, Jinsun Park

- *Asian Conference on Computer Vision (ACCV)*, 2024. [\[Paper\]](#)

UNDER REVIEW

4. OceanSplat: Object-aware Gaussian Splatting with Trinocular View Consistency for Underwater Scene Reconstruction

Minseong Kweon, Jinsun Park

- *Under review* (double-blind)

3. CAR-Stereo: Confidence-aware Adaptive Disparity Refinement for Real-time Stereo Matching

Chanill Park, Janghyun Kim, Minseong Kweon, Jinsun Park

- *Under review* (double-blind)

2. MrGS: Multi-modal Radiance Fields with 3D Gaussian Splatting for RGB-Thermal Novel View Synthesis

Minseong Kweon, Janghyun Kim, Ukcheol Shin, Jinsun Park

- *Working-in-progress* [\[Project Page\]](#)

* Short version at Thermal Infrared Robotics (TIRO) Workshop, ICRA 2025. [\[Poster\]](#)

* Received **Best Poster Award**.

1. All-day Depth Completion via Thermal-LiDAR Fusion

Janghyun Kim, Minseong Kweon, Jinsun Park, Ukcheol Shin

- *Under review* at IEEE Transactions on Intelligent Vehicles (T-IV). [\[Paper\]](#)

* Short version at Thermal Infrared Robotics (TIRO) Workshop, ICRA 2025. [\[Poster\]](#)

AWARDS & HONORS

- **Best Poster Award**, Thermal Infrared Robotics (TIRO) Workshop, ICRA 2025.
- **Honorable Mention**, Undergraduate AI Paper Competition, Summer Annual Conference of IEIE, 2024

ACADEMIC SERVICES

- **Program Committee & Reviewer** AAAI Conference on Artificial Intelligence (AAAI), 2026

RESEARCH EXPERIENCES

VGG-Group, University of Science and Technology of China

Visiting Research Student

Summer 2024

Hefei, China

Advisor: Dr. Xuejin Chen

Research Area: 3D Reconstruction

Project: Distributed 3D Gaussian Splatting for large-scale scene representation

- Developed a two-stage 3D Gaussian Splatting framework (multi-GPU densification + single-GPU fine-tuning) for large-scale 3D reconstruction, thereby enabling the optimization of over 10M primitives to cover areas spanning several km².
- Proposed a distributed 3DGS framework with adaptive positional noise to enhance rendering quality in high-frequency regions, achieving +0.33 dB PSNR improvement and ×3 faster training on *MatrixCity* dataset compared to CityGS.

Related Publication: “NoiseGS: Boosting 3D Gaussian Splatting with Positional Noise for Large-Scale Scene Rendering”, Eurographics (short paper) 2025.

Mobility-Robot Research Division, Electronics and Telecommunications Research Institute

Research Intern

Summer 2023

Daejeon, Republic of Korea

Advisor: Dr. Seungmin Choi

Research Area: Visual Localization, Image Retrieval

Project: Visual place recognition for quadruped autonomous robots

- Developed hierarchical visual localization pipeline to achieve robust 6DoF pose estimation for quadruped robots.
- Developed mobile localization application using Unity and NAVER ARC eye API.
- Designed image retrieval module fusing local and global features in visual localization system, reducing the median pose error by 0.27° on the Aachen Day-Night dataset without re-ranking.

Related Publication: “ULTRON: Unifying Local Transformer and Convolution for Large-Scale Image Retrieval”, ACCV 2024.

WORK EXPERIENCES

Sandburg Inc.

Software Engineer

Jun. 2022 – Dec. 2022

Busan, Republic of Korea

- Developed a sales-based credit prediction machine learning module for Alternative Credit Scoring System (ACSS).

PRESENTATIONS

3. Self-Similarity Feature Learning for Person Re-Identification
- Minseong Kweon, Jinsun Park
- Summer Annual Conference of Institute of Electronics and Information Engineers (IEIE), 2024. [\[Paper \(in Korean\)\]](#)
 - * Received **Honorable Mention** at Undergraduate AI Paper Competition.
2. Hierarchical Visual Localization for Indoor Autonomous Systems
- Minseong Kweon, Jeonghyun Noh, Chanill Park, Jinsun Park
- Korea Robotics Conferences (KRoC), 2024. [\[Paper \(in Korean\)\]](#)
1. Attention-based Human Activity Recognition with 3-axis Accelerometer Data Conversion
- Minseong Kweon, Jaehyeong Park, Kyunghyun Kim, Jeonghyun Noh, Jinsun Park
- Korea Computer Congress (KCC), 2023. [\[Paper \(in Korean\)\]](#)

PATENTS

1. Minseong Kweon, Jinsun Park, “METHOD AND SYSTEM FOR PROCESSING IMAGES BASED ON LOCAL ATTENTION”. Korean Patent Application No. 10-2024-0176234. Patent Pending.

SKILLS

Languages	C/C++, Python, MATLAB, JavaScript
Tools	Github, Linux, Wandb, ROS
Libraries	NumPy, SciPy, Pytorch, Pytorch3D, OpenCV, Open3D, PyBullet, PyRender, scikit-learn